Syllabus Details

Syllabus ID:	13892
Syllabus Name:	**Advanced Cross-Platform Application Programming With .NET_Lập trình ứng dụng đa nền tảng nâng cao với .NET**
Course Name English:	**Advanced Cross-Platform Application Programming With .NET**
Subject Code:	**PRN222**
Learning-Teaching Method:	Social Constructivism,In-class lecture,Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 49.5h contact hours + 145' Final exam + 98.1h self-study
Pre-Requisite:	PRN212
Description:	Upon completion of this course students should: 1. Understand the followings: • Apply C# language for developing tranditional and realtime communication Web applications; • Fundamental concepts of .NET Core Platform • Basic knowledge of ASP.NET Core with Realtime Communication using SignalR • Basic knowledge of ASP.NET Core MVC, Razor Pages, Blazor with tranditional Web App and apply into ASP.NET Core application • Basic knowledge of Asynchronous and Parallel Programming in .NET Core application • Basic knowledge of Dependency Injection apply into .NET Core applications • Basic knowledge of Worker Service and apply to implement Background Tasks 2. Be able to: • Develop Cross - platform Web applications by ASP.NET Core MVC, Razor Pages, Blazor • Implement Real-time applications by Signal R and ASP.NET Core • Implement Background Tasks with Worker Service 3. Be able to work in a team and present group's results
StudentTasks:	- Students must attend at least 80% of contact sessions in order to be accepted to the final examination. - Student is responsible to do all assigned exercises given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Promptly access to the FU CMS at http://flm.fpt.edu.vn for up-to-date course information
Tools:	- Internet - Visual Studio.NET 2022 or later - MS SQL Server 2019 or later - Platform : .NET 8.0 or later
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	1402/QĐ-ĐHFPT dated 12/12/2025
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	12/12/2025

5 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	https://learn.microsoft.com/en-us/ef/core/	Microsoft	Microsoft	N/A	N/A		True	False	True	
2	https://learn.microsoft.com/en-us/dotnet/core/introduction	Microsoft	Microsoft	N/A	N/A		True	False	True	

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
3	https://learn.microsoft.com/en-us/dotnet/csharp/whats-new/csharp-12	Microsoft	Microsoft	N?A			True	False	True	
4	C# 12 in a Nutshell	Joseph Albahari	O'Reilly Media, Inc.	2024	N/A	9781098147440	False	True	False	
5	C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals	Mark J. Price	Packt Publishing Ltd.	2023	8th Edition	9781837635870	False	True	False	

5 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Demonstrate about Asynchronous & Parallel programming ; Dependency Injection
2	CLO2	Demonstrate about Model-View-Controller in ASP.NET Core
3	CLO3	Demonstrate about Razor Pages in ASP.NET Core
4	CLO4	Demonstrate about Web Application with Blazor
5	CLO5	Demonstrate about Real-time communication and Worker Service

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Course Introduction	Offline	CLO1	I	Slide	PRN222	- Read the slide, related reference documents - Do exercises - Self practice	
2	Review C# basics and Entity Framework Core	Offline	CLO1	TU	Slide		- Read the slide, related reference documents - Do exercises - Self practice	
3	Group Project Introduction	Offline	CLO1	TU	Project List		- Choose Group (3-5 Students) - Choose Topic - Collect requirements about Topic	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
4	Chapter 01: Networking Programming	Offline	CLO1	TU	https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/overview https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/http/http-overview https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/sockets/sockets-overview		- Read the slide, related reference documents - Do exercises - Self practice	
5	Chapter 01: Networking Programming (cont.)	Offline	CLO1	TU	https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/overview https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/http/http-overview https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/sockets/sockets-overview		- Read the slide, related reference documents - Do exercises - Self practice	
6	Chapter 01: Networking Programming (cont.)	Offline	CLO1	TU	https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/overview https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/http/http-overview https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/sockets/sockets-overview		- Read the slide, related reference documents - Do exercises - Self practice	
7	Chapter 01: Networking Programming (cont.)	Offline	CLO1	TU	https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/overview https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/http/http-overview https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/sockets/sockets-overview		- Read the slide, related reference documents - Do exercises - Self practice	
8	Chapter 01: Networking Programming (cont.)	Offline	CLO1	TU	https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/overview https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/http/http-overview https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/sockets/sockets-overview		- Read the slide, related reference documents - Do exercises - Self practice	
9	Chapter 01: Networking Programming (cont.)	Offline	CLO1	TU	https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/overview https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/http/http-overview https://learn.microsoft.com/en-us/dotnet/fundamentals/networking/sockets/sockets-overview		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
10	Chapter 02: Asynchronous and Parallel Programming in .NET	Offline	CLO1	T: Constructivism Method	https://learn.microsoft.com/en-us/dotnet/standard/parallel-programming/ https://learn.microsoft.com/en-us/dotnet/csharp/asynchronous-programming/		- Read the slide, related reference documents - Do exercises - Self practice	
11	Chapter 02: Asynchronous and Parallel Programming in .NET (cont.)	Offline	CLO1	T: Constructivism Method	https://learn.microsoft.com/en-us/dotnet/standard/parallel-programming/ https://learn.microsoft.com/en-us/dotnet/csharp/asynchronous-programming/		- Read the slide, related reference documents - Do exercises - Self practice	
12	Chapter 02: Asynchronous and Parallel Programming in .NET (cont.)	Offline	CLO1	T: Constructivism Method	https://learn.microsoft.com/en-us/dotnet/standard/parallel-programming/ https://learn.microsoft.com/en-us/dotnet/csharp/asynchronous-programming/		- Read the slide, related reference documents - Do exercises - Self practice	
13	Project Assistant - 01	Offline	CLO1		Project List		- Discuss project problems	
14	Chapter 03: Dependency Injection in .NET	Offline	CLO1	T: Constructivism Method	https://learn.microsoft.com/en-us/dotnet/core/extensions/dependency-injection https://learn.microsoft.com/en-us/aspnet/core/fundamentals/dependency-injection?view=aspnetcore-8.0		- Read the slide, related reference documents - Do exercises - Self practice	
15	Chapter 03: Dependency Injection in .NET (cont.)	Offline	CLO1	T: Constructivism Method	https://learn.microsoft.com/en-us/dotnet/core/extensions/dependency-injection https://learn.microsoft.com/en-us/aspnet/core/fundamentals/dependency-injection?view=aspnetcore-8.1		- Read the slide, related reference documents - Do exercises - Self practice	
16	Chapter 03: Dependency Injection in .NET (cont.)	Offline	CLO1	T: Constructivism Method	https://learn.microsoft.com/en-us/dotnet/core/extensions/dependency-injection https://learn.microsoft.com/en-us/aspnet/core/fundamentals/dependency-injection?view=aspnetcore-8.2		- Read the slide, related reference documents - Do exercises - Self practice	
17	Chapter 03: Dependency Injection in .NET (cont.)	Offline	CLO1	T: Constructivism Method	https://learn.microsoft.com/en-us/dotnet/core/extensions/dependency-injection https://learn.microsoft.com/en-us/aspnet/core/fundamentals/dependency-injection?view=aspnetcore-8.3		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
18	Chapter 03: Dependency Injection in .NET (cont.)	Offline	CLO1	T: Constructivism Method	https://learn.microsoft.com/en-us/dotnet/core/extensions/dependency-injection https://learn.microsoft.com/en-us/aspnet/core/fundamentals/dependency-injection?view=aspnetcore-8.4		- Read the slide, related reference documents - Do exercises - Self practice	
19	Chapter 04: Building Web Application using ASP.NET Core MVC	Offline	CLO2	TU	https://learn.microsoft.com/vi-vn/aspnet/core/tutorials/first-mvc-app/start-mvc?view=aspnetcore-8.0 Chapter 12. Introducing Web Development Using ASP.NET Core (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
20	Chapter 04: Building Web Application using ASP.NET Core MVC (cont.)	Offline	CLO2	TU	https://learn.microsoft.com/vi-vn/aspnet/core/tutorials/first-mvc-app/start-mvc?view=aspnetcore-8.0 Chapter 12. Introducing Web Development Using ASP.NET Core (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
21	Chapter 04: Building Web Application using ASP.NET Core MVC (cont.)	Offline	CLO2	TU	https://learn.microsoft.com/vi-vn/aspnet/core/tutorials/first-mvc-app/start-mvc?view=aspnetcore-8.0 Chapter 12. Introducing Web Development Using ASP.NET Core (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
22	Chapter 04: Building Web Application using ASP.NET Core MVC (cont.)	Offline	CLO2	TU	https://learn.microsoft.com/vi-vn/aspnet/core/tutorials/first-mvc-app/start-mvc?view=aspnetcore-8.0 Chapter 12. Introducing Web Development Using ASP.NET Core (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
23	Chapter 04: Building Web Application using ASP.NET Core MVC (cont.)	Offline	CLO2	TU	https://learn.microsoft.com/vi-vn/aspnet/core/tutorials/first-mvc-app/start-mvc?view=aspnetcore-8.0 Chapter 12. Introducing Web Development Using ASP.NET Core (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
24	Chapter 04: Building Web Application using ASP.NET Core MVC (cont.)	Offline	CLO2	TU	https://learn.microsoft.com/vi-vn/aspnet/core/tutorials/first-mvc-app/start-mvc?view=aspnetcore-8.0 Chapter 12. Introducing Web Development Using ASP.NET Core (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals)		- Read the slide, related reference documents - Do exercises - Self practice	
25	Lab 01	Offline	CLO2	TU	Lab 01		- Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
26	Assignment 01	Offline	CLO2	TU	Assignment 01		- Discuss assignment problems	
27	Progress Test 01	Offline	CLO2	TU	LMS		- Do Quiz	
28	Chapter 05: Building Websites Using ASP.NET Core Razor Pages	Offline	CLO3	TU	Chapter 13. Building Websites Using ASP.NET Core Razor Pages (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/tutorials/razor-pages/?view=aspnetcore-8.0		- Read the slide, related reference documents - Do exercises - Self practice	
29	Chapter 05: Building Websites Using ASP.NET Core Razor Pages (cont.)	Offline	CLO3	TU	Chapter 13. Building Websites Using ASP.NET Core Razor Pages (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/tutorials/razor-pages/?view=aspnetcore-8.1		- Read the slide, related reference documents - Do exercises - Self practice	
30	Chapter 05: Building Websites Using ASP.NET Core Razor Pages (cont.)	Offline	CLO3	TU	Chapter 13. Building Websites Using ASP.NET Core Razor Pages (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/tutorials/razor-pages/?view=aspnetcore-8.2		- Read the slide, related reference documents - Do exercises - Self practice	
31	Chapter 05: Building Websites Using ASP.NET Core Razor Pages (cont.)	Offline	CLO3	TU	Chapter 13. Building Websites Using ASP.NET Core Razor Pages (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/tutorials/razor-pages/?view=aspnetcore-8.3		- Read the slide, related reference documents - Do exercises - Self practice	
32	Chapter 05: Building Websites Using ASP.NET Core Razor Pages (cont.)	Offline	CLO3	TU	Chapter 13. Building Websites Using ASP.NET Core Razor Pages (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/tutorials/razor-pages/?view=aspnetcore-8.4		- Read the slide, related reference documents - Do exercises - Self practice	
33	Chapter 05: Building Websites Using ASP.NET Core Razor Pages (cont.)	Offline	CLO3	TU	Chapter 13. Building Websites Using ASP.NET Core Razor Pages (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/tutorials/razor-pages/?view=aspnetcore-8.5		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
34	Chapter 05: Building Websites Using ASP.NET Core Razor Pages (cont.)	Offline	CLO3	TU	Chapter 13. Building Websites Using ASP.NET Core Razor Pages (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/tutorials/razor-pages/?view=aspnetcore-8.6		- Read the slide, related reference documents - Do exercises - Self practice	
35	Chapter 05: Building Websites Using ASP.NET Core Razor Pages (cont.)	Offline	CLO3	TU	Chapter 13. Building Websites Using ASP.NET Core Razor Pages (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/tutorials/razor-pages/?view=aspnetcore-8.7		- Read the slide, related reference documents - Do exercises - Self practice	
36	Chapter 05: Building Websites Using ASP.NET Core Razor Pages (cont.)	Offline	CLO3	TU	Chapter 13. Building Websites Using ASP.NET Core Razor Pages (C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/tutorials/razor-pages/?view=aspnetcore-8.8		- Read the slide, related reference documents - Do exercises - Self practice	
37	Chapter 06: Building a Web App with Blazor and ASP .Net Core	Offline	CLO4	TU	Chapter 15. Building User Interfaces Using Blazor ((C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/blazor/tutorials/?view=aspnetcore-8.0		- Read the slide, related reference documents - Do exercises - Self practice	
38	Chapter 06: Building a Web App with Blazor and ASP .Net Core (cont.)	Offline	CLO4	TU	Chapter 15. Building User Interfaces Using Blazor ((C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/blazor/tutorials/?view=aspnetcore-8.1		- Read the slide, related reference documents - Do exercises - Self practice	
39	Chapter 06: Building a Web App with Blazor and ASP .Net Core (cont.)	Offline	CLO4	TU	Chapter 15. Building User Interfaces Using Blazor ((C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/blazor/tutorials/?view=aspnetcore-8.2		- Read the slide, related reference documents - Do exercises - Self practice	
40	Chapter 06: Building a Web App with Blazor and ASP .Net Core (cont.)	Offline	CLO4	TU	Chapter 15. Building User Interfaces Using Blazor ((C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/blazor/tutorials/?view=aspnetcore-8.3		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
41	Chapter 06: Building a Web App with Blazor and ASP .Net Core (cont.)	Offline	CLO4	TU	Chapter 15. Building User Interfaces Using Blazor ((C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/blazor/tutorials/?view=aspnetcore-8.4		- Read the slide, related reference documents - Do exercises - Self practice	
42	Chapter 06: Building a Web App with Blazor and ASP .Net Core (cont.)	Offline	CLO4	TU	Chapter 15. Building User Interfaces Using Blazor ((C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/blazor/tutorials/?view=aspnetcore-8.5		- Read the slide, related reference documents - Do exercises - Self practice	
43	Chapter 06: Building a Web App with Blazor and ASP .Net Core (cont.)	Offline	CLO4	TU	Chapter 15. Building User Interfaces Using Blazor ((C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/blazor/tutorials/?view=aspnetcore-8.6		- Read the slide, related reference documents - Do exercises - Self practice	
44	Chapter 06: Building a Web App with Blazor and ASP .Net Core (cont.)	Offline	CLO4	TU	Chapter 15. Building User Interfaces Using Blazor ((C# 12 and .NET 8 – Modern Cross-Platform Development Fundamentals) https://learn.microsoft.com/en-us/aspnet/core/blazor/tutorials/?view=aspnetcore-8.7		- Read the slide, related reference documents - Do exercises - Self practice	
45	Project Assistant - 02	Offline	CLO4	TU	Project List		- Discuss project problems	
46	Chapter 07: Real-Time Communication	Offline	CLO5	TU	https://learn.microsoft.com/en-us/aspnet/core/signalr/introduction?view=aspnetcore-8.0 https://dotnet.microsoft.com/en-us/apps/aspnet/signalr		- Read the slide, related reference documents - Do exercises - Self practice	
47	Chapter 07: Real-Time Communication (cont.)	Offline	CLO5	TU	https://learn.microsoft.com/en-us/aspnet/core/signalr/introduction?view=aspnetcore-8.0 https://dotnet.microsoft.com/en-us/apps/aspnet/signalr		- Read the slide, related reference documents - Do exercises - Self practice	
48	Chapter 07: Real-Time Communication (cont.)	Offline	CLO5	TU	https://learn.microsoft.com/en-us/aspnet/core/signalr/introduction?view=aspnetcore-8.0 https://dotnet.microsoft.com/en-us/apps/aspnet/signalr		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
49	Chapter 07: Real-Time Communication (cont.)	Offline	CLO5	TU	https://learn.microsoft.com/en-us/aspnet/core/signalr/introduction?view=aspnetcore-8.0 https://dotnet.microsoft.com/en-us/apps/aspnet/signalr		- Read the slide, related reference documents - Do exercises - Self practice	
50	Chapter 07: Real-Time Communication (cont.)	Offline	CLO5	TU	https://learn.microsoft.com/en-us/aspnet/core/signalr/introduction?view=aspnetcore-8.0 https://dotnet.microsoft.com/en-us/apps/aspnet/signalr		- Read the slide, related reference documents - Do exercises - Self practice	
51	Chapter 07: Real-Time Communication (cont.)	Offline	CLO5	TU	https://learn.microsoft.com/en-us/aspnet/core/signalr/introduction?view=aspnetcore-8.0 https://dotnet.microsoft.com/en-us/apps/aspnet/signalr		- Read the slide, related reference documents - Do exercises - Self practice	
52	Lab 02	Offline	CLO5	TU	Lab 02		- Do exercises - Self practice	
53	Assignment 02	Offline	CLO5	TU	Assignment 02		- Discuss assignment problems	
54	Progress Test 02	Offline	CLO5	TU	LMS		- Do Quiz	
55	Chapter 08: Implement Background Tasks with Worker Service	Offline	CLO5	TU	https://learn.microsoft.com/en-us/aspnet/core/fundamentals/host/hosted-services?view=aspnetcore-8.0		- Read the slide, related reference documents - Do exercises - Self practice	
56	Chapter 08: Implement Background Tasks with Worker Service (cont.)	Offline	CLO5	TU	https://learn.microsoft.com/en-us/aspnet/core/fundamentals/host/hosted-services?view=aspnetcore-8.1		- Read the slide, related reference documents - Do exercises - Self practice	
57	Chapter 08: Implement Background Tasks with Worker Service (cont.)	Offline	CLO5	TU	https://learn.microsoft.com/en-us/aspnet/core/fundamentals/host/hosted-services?view=aspnetcore-8.2		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
58	Chapter 08: Implement Background Tasks with Worker Service (cont.)	Offline	CLO5	TU	https://learn.microsoft.com/en-us/aspnet/core/fundamentals/host/hosted-services?view=aspnetcore-8.3		- Read the slide, related reference documents - Do exercises - Self practice	
59	Chapter 08: Implement Background Tasks with Worker Service (cont.)	Offline	CLO5	TU	https://learn.microsoft.com/en-us/aspnet/core/fundamentals/host/hosted-services?view=aspnetcore-8.4		- Read the slide, related reference documents - Do exercises - Self practice	
60	Chapter 08: Implement Background Tasks with Worker Service (cont.)	Offline	CLO5	TU	https://learn.microsoft.com/en-us/aspnet/core/fundamentals/host/hosted-services?view=aspnetcore-8.5		- Read the slide, related reference documents - Do exercises - Self practice	
61	Project Evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	U			Group Presentation	
62	Project Evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO6	U			Group Presentation	
63	Project Evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO7	U			Group Presentation	
64	Project Evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO8	U			Group Presentation	
65	Project Evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO9	U			Group Presentation	
66	Project Evaluation	Offline	CLO1, CLO2, CLO3, CLO4, CLO10	U			Group Presentation	

0 Constructive question(s)

6 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment	on-going	2	10.0%	>0	12 slots	Assignment 1 CLO2, CLO3 Assignment 1 CLO4			Subjects in syllabus		
2	Group Project	on-going	1	25.0%	>0	20 slots	CLO2, CLO3, CLO4, CLO5			Subjects in syllabus		
3	Progress test	on-going	2	10.0%	>0	30'	PT1: CLO1, CLO2, CLO3 PT1: CLO4, CLO5	Multiple choices Marked by Computer or a suitable format	20/each			
4	Final exam	Final exam	2	55.0%		145'	All CLOs			All subjects in syllabus	By exam board Manual evaluating	TE >= 4, PE >0, & Min Avg Mark To Pass >= 5
5	Practical Exam	Final exam	1	25.0%	>0	85'	problem(s) to solve by programming using CLO1 - CLO5			Subjects in syllabus	By exam board Manual evaluating	
6	Theoretical exam	Final exam	1	30.0%	4	60'	questions distribution:CLO1.x: 5 - 10CLO2.x: 5 - 10CLO3.x: 5 - 10CLO4.x: 5 - 10CLO4.x: 5 - 10		50	All subjects in syllabus	By exam board	